Karl Weisenburger

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U.S. Citizen

EDUCATION

Ph.D. Candidate in Mathematics (Computational Science) **Aug 2022 – Present**
Purdue University (West Lafayette, IN)

Advisors: Gregery Buzzard & Charles Bouman

Areas: Bayesian inverse problems, nuisance parameter estimation, tomography

Relevant Coursework: Deep Learning; Computational Optimal Transport & Generative Models; Digital Image Processing I & II; Diffraction, Fourier Optics & Imaging; Numerical Analysis

Expected: Spring 2027

B.S. in Applied Mathematics, Minor in Latin **2018 – 2022**
Hillsdale College (Hillsdale, MI)

Honors: Summa Cum Laude, GPA 3.95/4.0

RESEARCH EXPERIENCE

Graduate Research Assistant **May 2023 – Present**
Purdue University (West Lafayette, IN)

- Developed **WindDensity-MBIR**, a model-based iterative reconstruction method that recovers high-order wavefront features to within 10–25% error in severely underdetermined regimes (sparse views, narrow field of view, limited angular extent), even with piston, tip, and tilt removed from the measurements.
- Extended the group's JAX-based MBIR software package to support multi-axis tomography geometries.

Doctoral Research Fellow (ORISE) **May – Aug 2026**
Air Force Institute of Technology (Virtual)

Mentor: Matthew Kemnetz

- Developed a single-shot multi-beam method for estimating unmeasured wavefront tip and tilt from higher-order wavefront data.
- In simulations, the estimated tip/tilt explained 66–78% of true tip/tilt coefficient variance.
- Correcting wavefronts with these estimates undid 78–85% of the upward shift in spectral peak frequency caused by the absence of tip/tilt and brought the spectral peak level to within 0.1 dB of the truth.

AFRL Scholars Program Summer Intern **Summers 2024, 2025**
Air Force Research Laboratory (Virtual)

Mentor: Matthew Kemnetz

- Built simulation and reconstruction software for tomographic wavefront sensing, simulating boiling-flow atmospheric turbulence and wavefront measurements through it.
- Explored the trade space of wavefront sensor configurations to guide experimental designs for density reconstructions.

Data Analytics Intern **May – Aug 2022**
Build Central (Virtual)

- Trained machine-learning models to predict construction project completion dates and built analytics dashboards used by leadership to productize the results.

PUBLICATIONS

Refereed Journal

- **K. J. Weisenburger**, G. T. Buzzard, C. A. Bouman, M. R. Kemnetz, "WindDensity-MBIR: Model-based iterative reconstruction for wind tunnel 3D density estimation," *Optical Engineering* **65**(8), 081820 (2026). doi:10.1117/1.OE.65.8.081820

Conference Proceedings

- **K. J. Weisenburger**, G. T. Buzzard, C. A. Bouman, M. R. Kemnetz, "Tomographic wavefront sensing simulations for wind tunnel measurements," *Proc. SPIE 13619, Unconventional Imaging, Sensing, and Adaptive Optics 2025*, 136190R (2025). doi:10.1117/12.3063657

SOFTWARE

WindDensity-MBIR — open-source Python implementation of the model-based iterative reconstruction method for wind-tunnel 3D density estimation; reproduces all results and figures from the *Optical Engineering* paper.

Code: `github.com/Karl-Weisenburger/WindDensity-MBIR`
Docs: `winddensity-mbir.readthedocs.io`

PRESENTATIONS

- *Model-Based Iterative Reconstruction for Wind Tunnel 3D Wavefront Estimation*, DE Student Education Workshop, Directed Energy Professional Society (DEPS) Annual DE Science & Technology Symposium (Mar–Apr 2026).
- *Wind Tunnel Tomographic Wavefront Sensing Simulations*, IS&T Electronic Imaging Symposium, Burlingame, CA (Mar 2026).
- *Tomographic Wavefront Sensing Simulations for Wind Tunnel Measurements*, SPIE Optics + Photonics, San Diego, CA (Aug 2025); Directed Energy Professional Society (DEPS) S&T Symposium (May 2025).

TECHNICAL SKILLS

Methods: Bayesian inverse problems, signal processing, nuisance parameter estimation, model-based iterative reconstruction (MBIR), optical system modeling and simulation, wavefront sensing.

Proficient: Python (JAX, NumPy, SciPy), MATLAB, LaTeX, Git.

Familiar: C++, PyTorch.

GPU & Parallel Computing: JAX vectorization (`vmap`) and JIT compilation for GPU-accelerated numerical code; multi-GPU sharding of parameter sweeps; Slurm on Purdue GPU clusters.

LEADERSHIP & SERVICE

President **2020 – 2021**
Hillsdale College Catholic Society (Hillsdale, MI)

- Recruited and managed three student boards; organized speaker events, lecture series, and retreats.

ESTEEM Mentor, *St. Thomas Aquinas Community Parish* **2025 – Present**